CS2600

THE UNIVERSITY OF WARWICK
Second Year Examinations, January 2017
Algorithms

- **Time allowed: 2 hours.**
- Answer **THREE** questions: **ONE** from Section A, and **TWO** from Section B.
- Make it clear on the cover of the answerbook which three questions you have selected.
- Calculators are not required and not permitted.

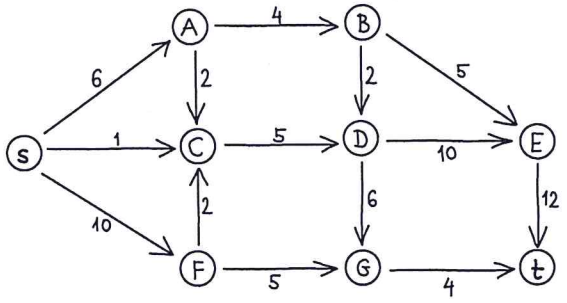
Continued

Section A

- Answer **ONE** question from this section.

Continued

1. (a) **[7 marks]** Consider the following flow network. The numbers labelling the edges are their capacities, and s and t are the distinguished *source* and *sink* vertices.



Give a flow of value 11 in the network.

State and justify the conservation constraint at vertex D and the capacity constraint at edge (B, E) for the flow that you have given.

(b) **[7 marks]** Give a maximum-value flow f in the network from part (a).

Draw the residual graph G_f, including residual capacities of all edges.

(c) **[6 marks]** Give a minimum-capacity s-t-cut $(X, V \backslash X)$ in the network from part (a).

Prove that the flow f is a maximum-value flow, and that the cut $(X, V \setminus X)$ is a minimum-capacity s-t-cut.

(d) **[10 marks]** The SQUARE search problem is—given a number x written in binary as input—to compute the binary representation of its square x^2. The MULTIPLY search problem is—given two numbers x and y written in binary as input—to compute the binary representation of their product xy.

Give a linear-time reduction from the MULTIPLY problem to the SQUARE problem.

Is it possible that squaring is algorithmically easier than multiplication, or in other words that SQUARE can be solved asymptotically faster than MULTIPLY?

Continued

2. In this problem we consider non-empty closed intervals: the interval $[s, f]$ "starts" at time s and "finishes" at time f, and hence we assume that $s < f$.

   (a) **[6 marks]** Two intervals $[s_1, f_1]$ and $[s_2, f_2]$ are *compatible* if they do not overlap, that is, if $f_1 < s_2$ (interval 1 finishes before interval 2 starts) or $f_2 < s_1$ (interval 2 finishes before interval 1 starts).

   Consider five intervals $[9, 11]$, $[2, 10]$, $[5, 8]$, $[3, 6]$, and $[1, 4]$. Draw the "incompatibility graph" for those intervals, in which there are five vertices $\{ 1, 2, 3, 4, 5 \}$, and there is an edge between vertices i and j if and only if intervals i and j are not compatible (that is, if they overlap).

   (b) **[6 marks]** Is $\{ 1, 3, 4 \}$ an independent set in the incompatibility graph from part (a)? Justify your answer.

   What is the largest independent set in the incompatibility graph from part (a)? Argue that it coincides with the largest set of intervals that are pairwise compatible.

   (c) **[6 marks]** Is $\{ 1, 2, 4 \}$ a vertex cover in the incompatibility graph from part (a)? Justify your answer.

   What is the smallest vertex cover in the incompatibility graph from part (a)?

   (d) **[12 marks]** The INTERVAL-SCHEDULING decision problem is—given n intervals $[s_1, f_1]$, $[s_2, f_2]$, ..., $[s_n, f_n]$, and given a number k—to determine if there are k intervals that are pairwise compatible.

   The VERTEX-COVER decision problem is—given an undirected graph $G = (V, E)$ and given a number ℓ—to determine if there is a vertex cover in G of size ℓ.

   Give a polynomial-time reduction from the INTERVAL-SCHEDULING problem to the VERTEX-COVER problem.

Continued

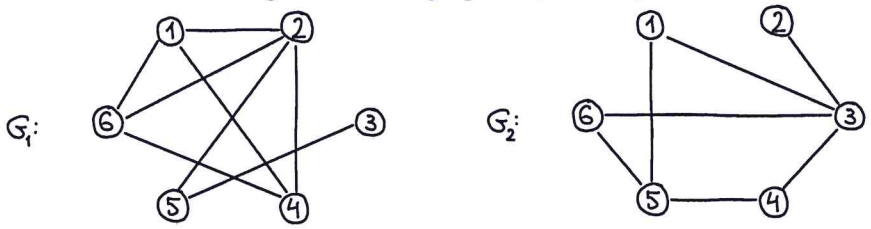
Section B

- Answer **TWO** questions from this section.

Continued

3. (a) **[5 marks]** A *clique* in an undirected graph $G = (V, E)$ is a subset $C \subseteq V$ of vertices in which every pair of vertices is connected by an edge. (In other words, $C \subseteq V$ is a clique if for all $v, w \in C$, we have $\{v, w\} \in E$.) The size of a clique C is defined to be the number of vertices in set C.

   Consider the following undirected graphs G_1 and G_2.



   What is the maximum size of a clique in graph G_1? Give a clique of maximum size in G_1.

   What is the maximum size of an independent set in graph G_2? Give an independent set of maximum size in G_2.

   (b) **[10 marks]** The SPARSE-SUBGRAPH decision problem is—given a graph G and numbers k and d—to determine if there is a set of k vertices of G such that there are at most d edges between them.

   Prove that the SPARSE-SUBGRAPH decision problem is in NP.

   (c) **[10 marks]** Recall that the INDEPENDENT-SET decision problem is—given a graph G and a number k—to determine if there is an independent set of size k in G.

   Design a polynomial-time reduction from the INDEPENDENT-SET decision problem to the SPARSE-SUBGRAPH decision problem.

   (d) **[10 marks]** The CLIQUE decision problem is—given a graph G and a number k—to determine if there is a clique of size k in G.

   Prove that the CLIQUE decision problem is NP-complete. You may assume that the INDEPENDENT-SET decision problem is NP-complete.

Continued

4. A *clique* in an undirected graph $G = (V, E)$ is a subset $C \subseteq V$ of vertices in which every pair of vertices is connected by an edge. The CLIQUE decision problem is—given a graph G and a number k as inputs—to decide if graph G contains a clique with k vertices as a subgraph.

   A *kite* is a graph with an even number of vertices, say $2k$, such that k of the vertices form a clique, and the other k vertices form a "tail" that is a path with k vertices, one end of which is connected by an edge to one of the vertices of the clique.

   (a) **[7 marks]** Draw a kite with 8 vertices.

   Then draw a graph with 10 vertices that contains a kite with 6 vertices as an induced subgraph (that is, when the other 4 vertices and their incident edges are removed from the graph then the remaining 6-vertex graph is a kite). Indicate the set of 3 vertices that form the clique in the kite, and the set of 3 vertices that form the tail of the kite.

   (b) **[14 marks]** The KITE decision problem is—given an undirected graph G and a number k as input—to decide if graph G contains a kite with $2k$ vertices as an induced subgraph.

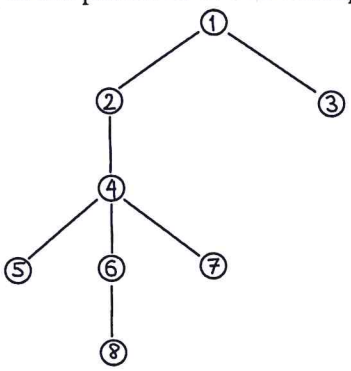
   Prove that KITE is in the complexity class NP by designing a polynomial-time certifier for the problem. Make it clear what the certificates are in your certifier, and how the certifier verifies that the certificate exhibits a kite.

   (c) **[14 marks]** Prove that KITE is NP-complete. You may assume that CLIQUE is NP-complete.

---

5. Recall that a set $C \subseteq V$ is a *vertex cover* of an undirected graph $G = (V, E)$ if for every edge $\{v, w\} \in E$ we have $v \in C$ or $w \in C$.

   Consider trees that are represented by a parent array p, such that if a vertex v is the root then $p[v] = v$, and otherwise $p[v]$ is the parent of v. For example, the following tree T:



   is represented by the parent array:

Vertex v	1	2	3	4	5	6	7	8
Parent $p[v]$ of v	1	1	1	2	4	4	4	6

   (a) **[4 marks]** Argue that $\{ 2, 3, 5, 6, 7 \}$ is a vertex cover of the example tree T above. Give a vertex cover of T of size 4.

   (b) **[6 marks]** What is the minimum size of a vertex cover of the example tree T above?

   Is there a minimum-size vertex cover of tree T, which contains the root vertex 1? If yes, give such a vertex cover; if not, explain why not.

   Is there a minimum-size vertex cover of tree T, which does not contain the root vertex 1? If yes, give such a vertex cover; if not, explain why not.

   (c) **[2 marks]** For a vertex v we let T_v be the subtree of tree T whose root is v.

   Draw the subtrees T_8, T_6, T_4, and T_2 of the example tree T above.

   (d) **[7 marks]** Define $M(v)$ to be the minimum size of a vertex cover of subtree T_v.

   Define $N(v)$ to be the minimum size of a vertex cover of subtree T_v that contains the root vertex v.

   For the example tree T above, find the values $N(v)$ and $M(v)$ for vertices $v = 8, 6, 4, 2$. In each case, give the minimum-size vertex covers of the relevant trees.

   (e) **[10 marks]** Describe an efficient polynomial-time algorithm that—given a tree as an input—computes the minimum size of a vertex cover of the tree.

   Justify the correctness of your algorithm. What is the asymptotic running time of your algorithm?

   (f) **[6 marks]** Design an linear-time algorithm that—given a rooted tree and the values $N(v)$ and $M(v)$ for all vertices v—computes a minimum-size vertex cover of the tree.

End